CS-301: Computer Networks Lab

Cisco Packet Tracer — Assignment 3

All practicals must be completed inside **Cisco Packet Tracer**.
Each assignment requires exactly **two screenshots** as submission.

Subject: CS-301 — Computer Networks
Tool: Cisco Packet Tracer 8.x or later

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_SetX.pkt` (e.g. `20221234_Set3.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of each set.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

DevNest Technologies has two departments in the same building but on completely separate networks: **Development** on the ground floor and **Testing** on the first floor. Because the two floors use different IP networks, a Development PC has no way to reach a Testing PC — every packet simply dies at the edge of its own subnet.

The network administrator, Ms. Laleh Ahmadi, has finally purchased a **router** and asks you to connect both floors through it so the two teams can communicate. Each floor will treat the router as its **default gateway**.

Device	IP Address	Default Gateway
Router Gi0/0	192.168.10.1	—
Dev-PC1	192.168.10.10	192.168.10.1
Dev-PC2	192.168.10.20	192.168.10.1
Dev-PC3	192.168.10.30	192.168.10.1
Router Gi0/1	192.168.20.1	—
Test-PC1	192.168.20.10	192.168.20.1
Test-PC2	192.168.20.20	192.168.20.1
Test-PC3	192.168.20.30	192.168.20.1
Subnet Mask: `255.255.255.0` on all PCs		

Your Task

- Place one router, two switches, and three PCs per floor. Connect each switch to its own router interface.
- Assign all IP addresses and default gateways from the table. Make sure both router interfaces are active before testing.
- Verify that a Development PC can ping a Testing PC across the router.

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View showing the router at the centre, both switches, and all six PCs connected with IP labels visible.
2. A **ping result** in the Command Prompt of any Development PC successfully reaching any Testing PC, showing zero packet loss.

Note: Router interfaces in Packet Tracer are **off by default**. Both interfaces must be turned on before traffic can pass through the router. Look for the Port Status setting inside the router's configuration panel.